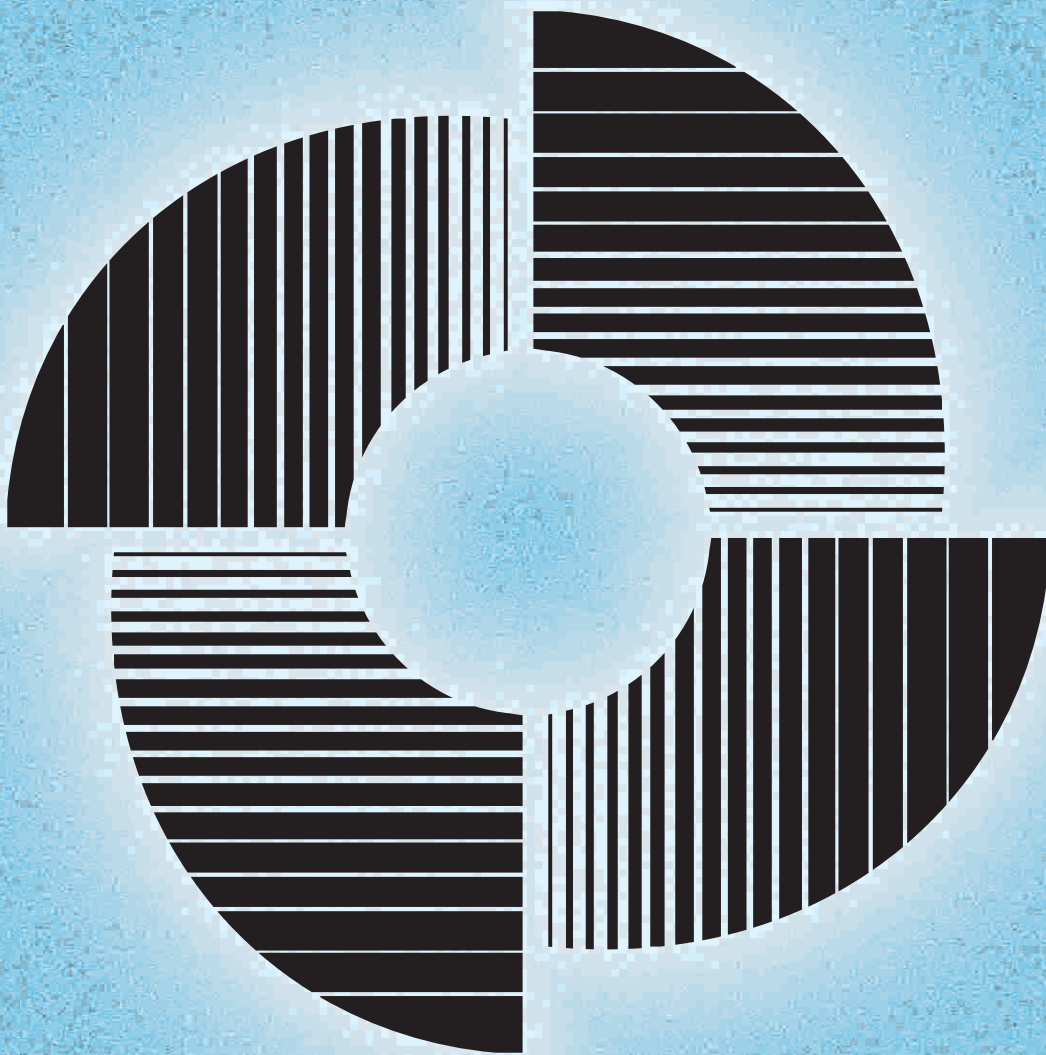


CENTRIFUGAL FANS & BLOWERS

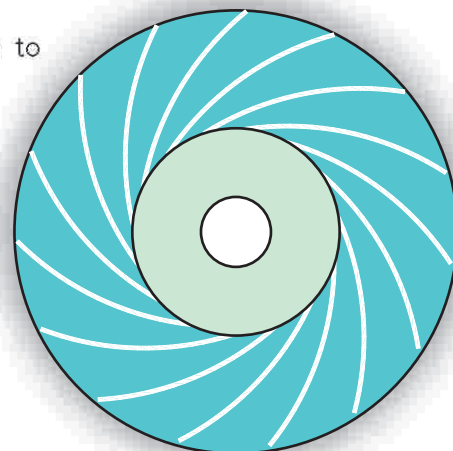


Durable • Cost effective • Energy efficient

High Performance Technology

Swam Centrifugal Fans & Blowers incorporate new concept in machinery design for most efficient performance in the air movement technology. The fans are manufactured in wide choice of sizes, arrangements and shapes of impellers to meet the air volume, pressure and temperature requirements for various types of industrial process applications. Ruggedly built, these are designed for trouble free continuous duty when operating under the most severe conditions required for material handling or for exhausting hot dirty fumes or gases.

To meet the quite rigorous requirements for the reliability of plants, our every effort is toward technological improvement in the areas of development, design, production, quality control and related services.



Design and Construction

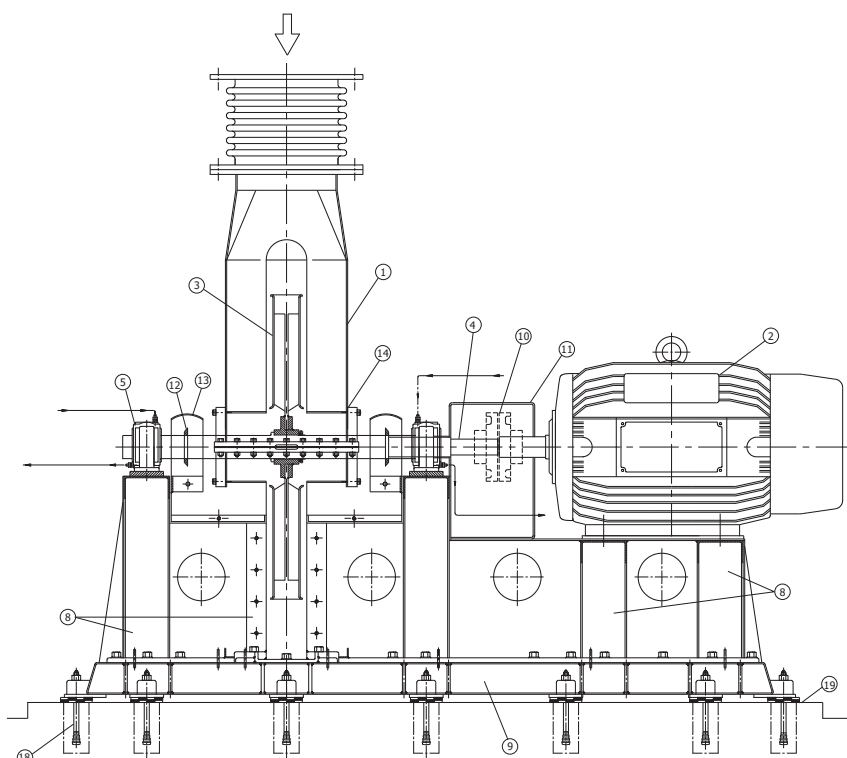
The Fans generally are of high quality welded construction and consist of Impeller assembly, spiral housing and suction and discharge nozzles. The impellers, depending on the application and the pressure, have the backward curved or straight radial blades.

The shaft sizes are selected based on critical speed consideration. The impellers of Fans intended to handle dust laden, corrosive or hot gases, are manufactured out of properly selected anti-corrosive, wear and heat resistant materials. The Impeller assembly is statically and dynamically balanced.

The Fan housing is ruggedly constructed, suitably reinforced to prevent deformation and vibration. It is of spiral shape, ideal for lower loss of dynamic pressure and resulting in high efficiency. The housing may have single or double suction nozzles and integral or with horizontal split.



S. No.	Description
19.	Antivibration Pads
18.	Foundation Anchor Fasteners
17.	Nuts & Bolts (At fan & Supports)
16.	Flange Gasket
15.	Fan casing Casket at Split Half
14.	Fan Casing Seal
13.	Deflector Guard
12.	Deflector Plate
11.	Coupling Guard
10.	Flexible Guard
9.	Base Plage
8.	Base Frame Box and Stiffners
7.	Expander (Discharge)
6.	Bellow (Suction / Discharge)
5.	Plummer Blocks
4.	Shaft
3.	Impeller
2.	Electric Motor
1.	Fan Casing



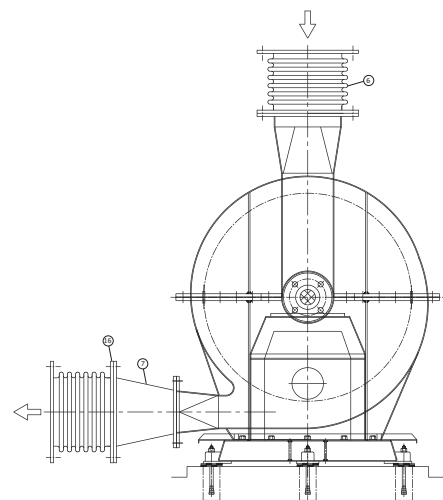
Impeller

The Fan impeller is generally fabricated by welding, rivetting or a combination of both, in appropriate profile. Considerations are light weight and ruggedness.

The material of construction would depend upon the air or gas being handled. Its temperature, dust content, abrasion and corrosive characteristic.

When handling abrasive gases with fine dust, the fitting of wear plates, hard facing or weld overlay are the standard measures.

A finished Impeller is dynamically balanced to grade 6.3 or 2.5 of ISO-1940 standard to ensure minimum vibration.



Impeller shaft

The fan drive shaft is normally machined from a carbon steel duly heat treated. However, for handling special gases, special alloy steels are used.

The critical speed of the shaft is calculated meticulously with due consideration of the moment of inertia of the impeller and the elasticity of bearings and base etc. The critical speed is kept more than 1.3 times the normal fan speed.

The fans handling special gases, the critical speed is adjusted according to the dust load and the operating temperature. The critical speed in such cases may be up to 1.4 times the normal speed with stability and safety aspects maintained.



Casing

The casing is of scroll like shape, for reducing the loss of dynamic pressure and thus resulting in high efficiency.

The casing is rugged, welded steel structure, sufficiently reinforced for preventing deformation, warp and vibration.

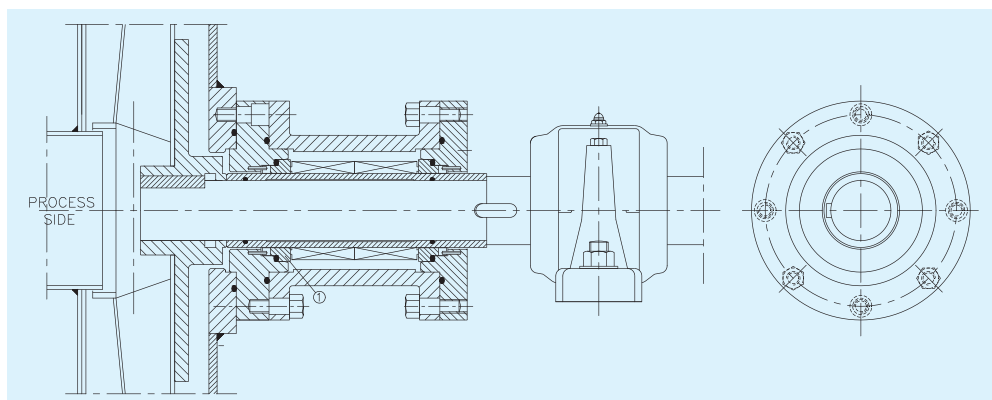
It is fitted with a suction cone, a drain on the bottom, and a manhole if required for easy inside inspection.

In large fans the casing is horizontally split for convenience of maintenance.

Fan handling high temperature gas, the base is suitably modified for the release of thermal stress caused by thermal expansion of the housing and to prevent its effects to the foundation.

Shaft Sealing

Generally asbestos packing are used, but labyrinth seals, mechanical seals etc., are also used in compliance with the condition of pressure, gas composition, and operating temperature.



Bearing and Lubrication System

The heavy duty ball and roller bearings are liberally sized based on fan speed, the radial and axial load, and for sufficient operating life. The bearings are fitted in the housing within specified tolerance to the drive shaft. The lubrication system is determined by the load conditions for bearings.

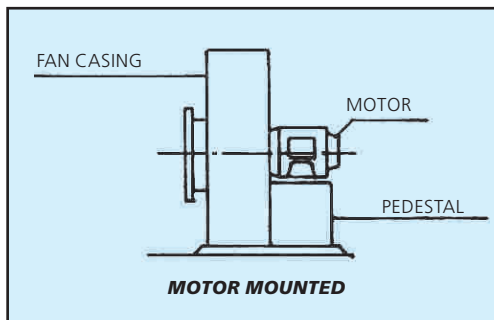
Ball and roller bearings are either self-lubricated with grease or by an automatic oil-bathing oil flinger, or lubricated by a forced oil system. The cooling methods of bearings are water and air cooling type, depending on the working conditions. Ball and roller bearings are cooled by heat flinger or water-through the jacket of the housing. Bearings are tightly sealed to keep out dust and to prevent oil leakage.

Drive Arrangements

There are various arrangements whereby the impeller of the fan can be driven. Each type has its distinct characteristic and is briefly discussed hereunder:-

Arrangement - M

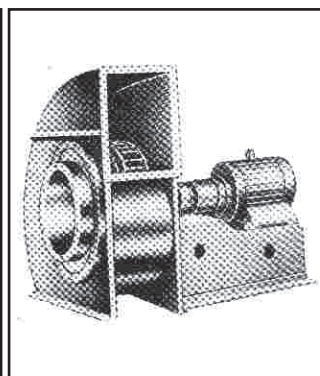
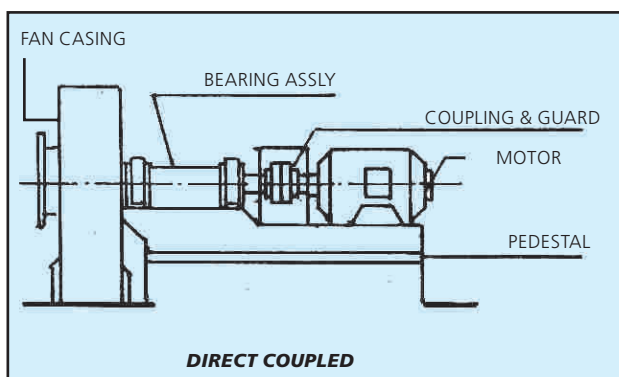
The Impeller is mounted directly on the motor shaft. This arrangement is preferred, when the fan speed corresponds to the motor speed and the temperature of the air or gas handled is not high.



- ◆ Compact size ◆ Economical ◆ Negligible drive losses ◆ Less maintenance ◆ Low installation cost.

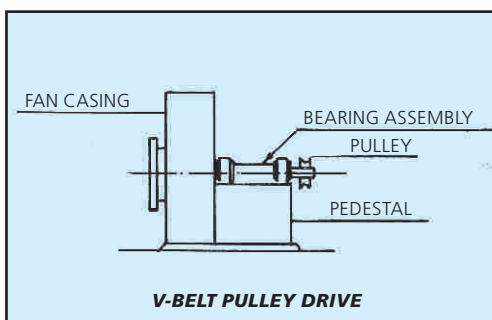
Arrangement - K

The impeller is directly coupled to motor shaft through a flexible coupling. This arrangement is preferred for large fans where it is not possible to mount the impellers direct on the motor shaft. The impeller is mounted at one end of shaft supported by the bearings on the other end.



Arrangement - R

The impeller shaft is mounted on two bearings and power is transmitted through V-belts and pulleys. The advantage of this arrangement is in its versatility and its ability to withstand higher operating temperature and flexibility to adjust the speeds to obtain exact flow and pressure.



Volume Control

The volume control and regulation is achievable by using the damper or by varying the blower operating speed.

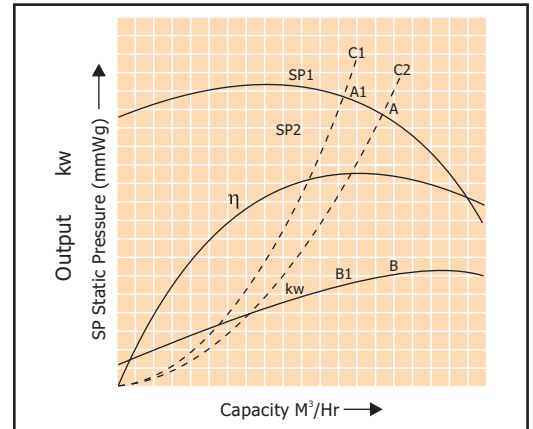
Three types of dampers; louver, vane control and butterfly type are applied for flow control both for volume and pressure. Depending on operating condition, a louver or butterfly damper is provided on either the inlet or the outlet of the fan and blower.

The following methods are applied.

Damper at outlet

A louver or butterfly valve can be provided on the outlet of a fan operated at rated speed. The air volume is regulated by the throttled effect. The resistance curve C1 shown in figure below is changed to C2 and the rated point A and B shift to A1 & B1.

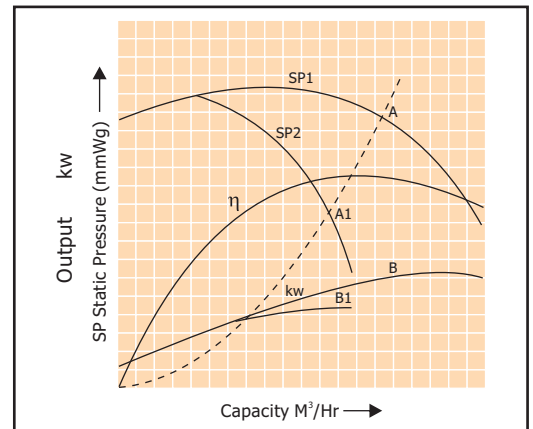
In case it is throttled below a certain point, surging will arise.



Damper at inlet

A louver or butterfly valve can be provided on the inlet of a fan operated at rated speed. The air volume is regulated by the throttled effect. The resistance curve C1 shown in figure below is changed to C2 and the rated point A and B shift to A1 & B1.

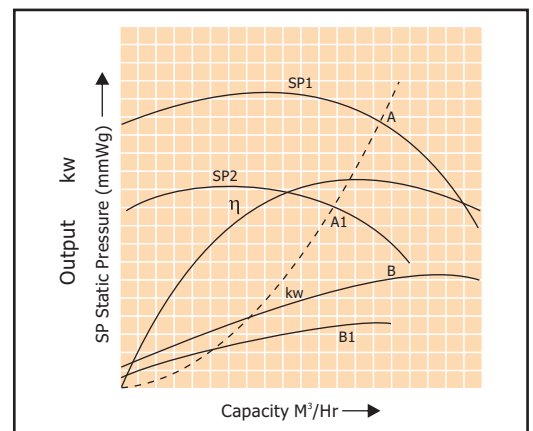
In case it is throttled below a certain point, surging will arise.



Flow Control by changing drive speed

Air volume is regulated by means of changing the Blower speed of through change of motor

The characteristic curves in such case gets changed as shown in figure. This is the most effective way of flow control for corresponding reduction in shaft power.

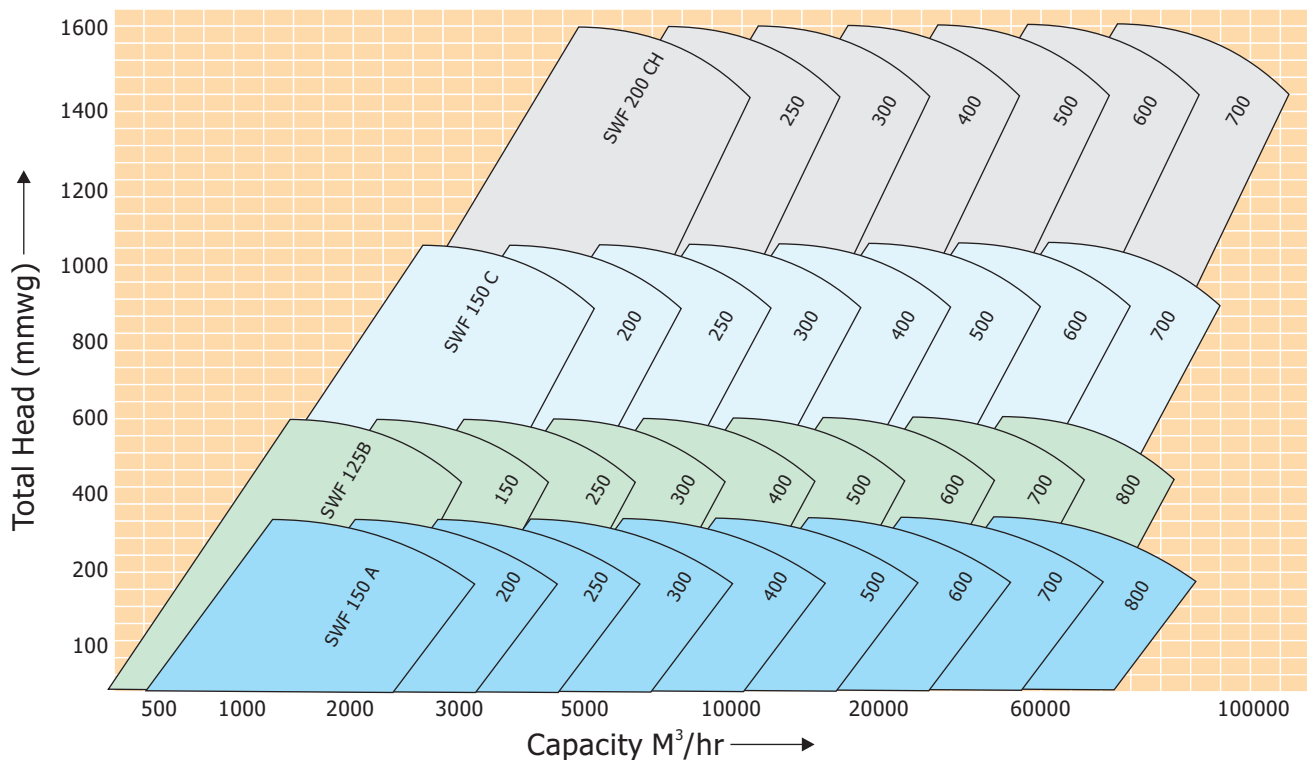


Material of Construction

For most common applications, the fans casing and impellers are fabricated out of carbon steel, but the fans are also supplied out of following materials based on customer requirement :

- ◆ Aluminium or Aluminum Alloys
- ◆ Stainless steel - 304, 316
- ◆ F.R.P., Rubber or P.V.C. lined
- ◆ Titanium etc.

PERFORMANCE SELECTION



The models indicated above show the relationship between total head & capacity and are designated with respect to the suction size.

For a specific fan characteristic - refer manufacturer.

Quality Control

The fan components and its materials are inspected to comply with specified standard and specification.

The quality assurance system encompasses self quality control by each worker supervised by the quality control engineer. In addition third party inspection is conducted for quality surveillance.

Welding quality is especially and scrupulously controlled and is subjected to a variety of tests, including the magnetic particle examination, radiographic test, dye penetrant test, aiming at to maintain a sound weld.

All measuring instruments used in both inspection and manufacturing checks are calibrated periodically.

The quality procedure are in compliance to ISO-9001 certification.

Assembly and Testing

The fan impeller is firmly keyed to shaft and is dynamically balanced with extreme precision to ensure quit and practically vibration free operation.

The assembly of fans or blowers is completed carefully and precisely in compliance to proper procedures. Thereafter a performance and continuous running test is performed. Bearing alignment is checked and adjusted if required.

During a min. two hour continuous running test at rated operating speed, the flow pressure, power vibration, bearing temperature, noise and other aspects are measured and tested to ensure trouble free operation on site.

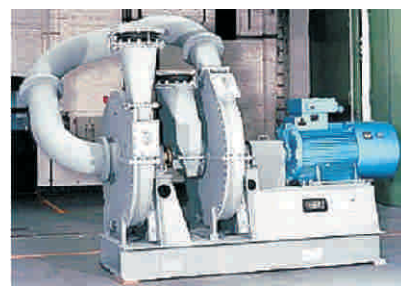
The performance test is carried out in compliance with IS-4894 or BS-848. Testing as per AMCA - 210 code is on special arrangement.

Special Application Fans

High pressure and high temperature fans

The high-pressure fans operate at high tip velocity and therefore, the impellers are required to be ruggedly built with relatively thick plate. The impellers are dynamically balanced to high precision.

The high temperature fans are fixed with deflector plate and are preferably oil lubricated.



Fans with Forced Lubrication

Depending on the severity of fan application, and its reliability of operation, the fan bearings are provided with forced lubrication. The arrangement is also provided for its cooling.

Double Suction Fans

Large capacity fans are provided with double suction. In such cases the Impeller is supported on bearings at two ends, and is located in the middle of the two bearing pedestals.



For Handling Hazardous, Toxic and Corrosive Gases

The fans are manufactured out of special materials suiting the characteristic of the gas being handled; such as stainless steel, carbon steel with zinc coating etc. The fans out of Titanium material and with mechanical seals are also offered when required.

Light Weight Fans

If the system requires the fans to be extra light weight and of compact size, the material of construction is Aluminium or its alloy.

The fans body is aluminium fabricated and the impellers is cast out of high grade aluminium alloy and then machined.

Performance Table for High Pressure Blowers
(Multi-Stage)

Model Designation	Nozzle Size (mm)		Capacity (CMH)		Pressure (mmwg)		Vacuum (mmhg)
	Suction	Discharge	Min.	Max.	Min.	Max.	Max.
SWF 200 CHM	200	200	1250	7000	1000	3000	1600
SWF 250 CHM	250	250	2500	12000	1000	3500	2000
SWF 300 CHM	300	300	4000	20000	1000	3500	2000
SWF 400 CHM	400	350	4700	28000	1200	4000	2000
SWF 500 CHM	500	400	9000	35000	1200	4500	2500
SWF 600 CHM	600	500	15000	50000	1500	5000	2500
SWF 750 CHM	750	600	22000	70000	1500	5000	2500

BLOWER MODEL DESIGNATION ILLUSTRATED : SWF 600CHMKG

SWF-SWAM CENTRIFUGAL FAN

600- SYMBOLIC SUCTION SIZE

CHM- DESIGN TYPE-HIGH PRESSURE MULTI STAGE

K - DIRECT COUPLED

G - GAS TIGHT SEAL



Applications

For most process industries; Cement, Power, Chemical, Steel & Mining Swam fans are used for :

- ◆ Ventilation & air circulation
- ◆ Induced draft & dust collection
- ◆ Cooling and combustion air
- ◆ Exhausting or Boosting Gases etc.

Distinctive features

- ◆ Wide selection of fans to meet customer process requirements
 - ◆ Heavy duty and rugged design for lasting performance
 - ◆ Smooth vibration free operation
 - ◆ Component post supply customer- support services
- ◆ Wide choice of materials of construction and sealing systems
- ◆ Suitable for corrosive, abrasive, hazardous application

OTHER PRODUCTS

Rotary T.L. Blowers
Rotary Dry Vacuum Pumps
Gas Boosters
Mechanical Vacuum Boosters
Industrial Silencers
Acoustic Hoods
Dust & Fume Extraction
Air Pressurisation Systems

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